CHAPTER 5

Surgical infection

Learning objectives

To understand:
- The characteristics of the common surgical pathogens and their sensitivities
- The factors that determine whether a wound will become infected
- The classification of sources of infection and their severity
- The clinical presentation of surgical infections
- The indications for and choice of prophylactic antibiotics
- The spectrum of commonly used antibiotics in surgery and principles of therapy
- The misuse of antibiotic therapy with the risk of resistance to antibiotics (such as methicillin-resistant *Staphylococcus aureus* [MRSA]) and emergence of resistant strains (such as *Clostridium difficile* enteritis)

To learn:
- Koch's postulates
- The management of abscesses
- The Surviving Sepsis Campaign, sepsis bundle and Sepsis Six
- Surgical implications of the COVID-19 pandemic

To appreciate:
- The importance of aseptic and antiseptic techniques and delayed primary or secondary closure in contaminated wounds

To be aware of:
- The causes of reduced resistance to infection (host response)

To know:
- The definitions of infection, particularly at surgical sites
- What basic precautions to take to avoid surgically relevant hospital-acquired infections

HISTORY OF SURGICAL INFECTION

Background

Surgical infections have always been a major complication related to surgery and trauma and have been documented for 4000–5000 years. Egyptians popularised some concepts about infection, as they were able to prevent putrefaction using their skills in mummification. Their medical papyruses also describe the use of salves and antiseptics to prevent surgical site infections (SSIs). This 'prophylaxis' had also been known earlier by the Assyrians, although it is less well documented. It was described again independently by the Greeks.

The Hippocratic teachings described the use of antimicrobials, such as wine and vinegar, which were widely used to irrigate open, infected wounds before delayed primary or secondary wound closure. A belief common to all these civilisations, and indeed even later to the Romans, was that whenever pus was localised in an infected wound it needed to be drained.

Galen recognised that this localisation of infection (suppuration) in wounds inflicted in the gladiatorial arena often heralded recovery, particularly after drainage (*pus bonum et laudabile*). Theodoric of Cervia, Ambroise Paré and Guy de Chauliac observed that clean wounds, closed primarily, could heal without infection or suppuration.

Koch's postulates

An understanding of the causes of infection came in the nineteenth century. Microbes had been seen under the microscope, but Koch laid down the first definition of infective disease (Koch's postulates; see ***Summary box 5.1***).

The Austrian obstetrician Ignaz Semmelweis showed that puerperal sepsis could be reduced from an incidence of over

Hippocrates of Kos, Greek physician and surgeon, and by common consent 'the father of medicine', was born on the island of Kos, off Turkey, about 460 BCE and probably died in 375 BCE.
Galen, 130–200, Roman physician, commenced practice as Surgeon to the Gladiators at Pergamum (now Bergama in Turkey) and later became personal physician to the Emperor Marcus Aurelius. As a prolific writer in anatomy, medicine, pathology and philosophy, his work affected medical thinking for 15 centuries after his death. (Gladiator is Latin for 'swordsman'.)
Theodoric of Cervia, 1210–1298, Bishop of Cervia, published a book on surgery ca. 1267.
Ambroise Paré, 1510–1590, French military surgeon, also worked at the Hotel Dieu, Paris, France.
Guy de Chauliac, 1298–1368, physician and chaplain to Pope Clement VI at Avignon, France, and the author of *Chirurgia Magna*, which was published about 1363.
Robert Koch, 1843–1910, Professor of Hygiene and Bacteriology, Berlin, Germany, stated his 'Postulates' in 1882.
Ignaz Semmelweis, 1818–1865, Professor of Obstetrics, Budapest, Hungary.

Summary box 5.1

Koch's postulates proving whether a given organism is the cause of a given disease

- It must be found in every case
- It should be possible to isolate it from the host and grow it in culture
- It should reproduce the disease when injected into another healthy host
- It should be recovered from an experimentally infected host

10% to under 2% by the simple act of handwashing between cases, particularly between postmortem examinations and the delivery suite.

Louis Pasteur recognised through his germ theory that microorganisms were responsible for infecting humans and causing disease. Joseph Lister applied this knowledge to the reduction of colonising organisms in compound fractures by using antiseptics. The principles of antiseptic surgery were soon enhanced with aseptic surgery at the turn of the twentieth century. As well as killing the bacteria on the skin before surgical incision (antiseptic technique), the conditions under which the operation was performed were kept free of bacteria (aseptic technique). This technique is still employed in modern operating theatres (*Figure 5.1*).

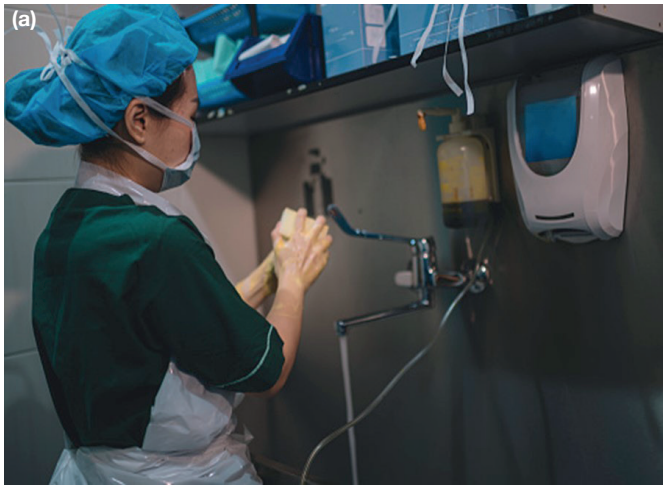
Discovery of antibiotics

The concept of a 'magic bullet' (*Zauberkugel*) that could kill microbes but not their host became a reality with the discovery of sulphonamide chemotherapy in the mid-twentieth century. The discovery of the antibiotic penicillin is attributed to Alexander Fleming in 1928, but it was not isolated for clinical use until 1941, by Florey and Chain. The first patient to receive penicillin was Police Constable Alexander in Oxford. Since then there has been a proliferation of antibiotics with broad-spectrum activity and antibiotics today remain the mainstay of antimicrobial therapy.

Misuse of antibiotic therapy with the risk of resistance

Many staphylococci today have become resistant to penicillin. Often bacteria develop resistance through the acquisition of β-lactamases, which break up the β-lactam ring present in the molecular structure of many antibiotics. The acquisition of extended-spectrum β-lactamases (ESBLs) is an increasing concern in some Gram-negative organisms that cause urinary tract infections because it is difficult to find an antibiotic effective against them. In addition, there is increasing concern about the rising resistance of many other bacteria to antibiotics, in particular the emergence of methicillin-resistant *Staphylococcus aureus* (MRSA), which is very relevant in general surgical practice.

The introduction of antibiotics for prophylaxis and for treatment, together with advances in anaesthesia and critical care medicine, has made possible surgery that would not previously have been considered. Faecal peritonitis is no longer inevitably fatal, and incisions made in the presence of such contamination can heal primarily without infection in over 90% of patients with appropriate antibiotic therapy. Despite



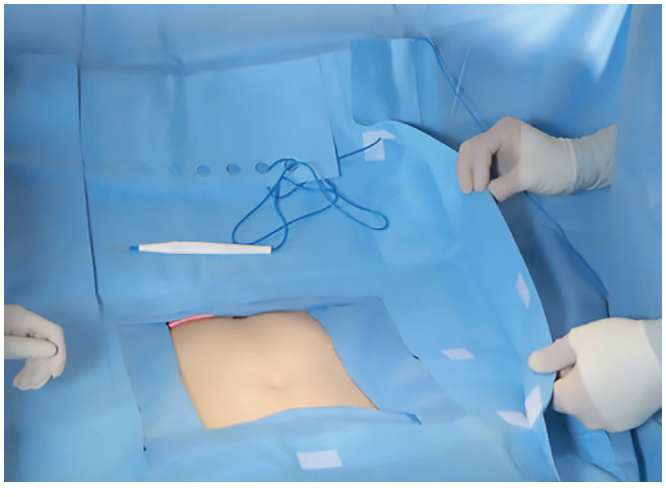


Figure 5.1 Aseptic techniques of scrubbing and draping in a modern operating theatre.

Louis Pasteur, 1822–1895, French chemist, bacteriologist and immunologist, Professor of Chemistry at the Sorbonne, Paris, France.
Lord Joseph Lister, 1827–1912, Professor of Surgery, Glasgow, Scotland (1860–1869), Edinburgh, Scotland (1869–1877) and King's College Hospital, London, England (1877–1892).
Sir Alexander Fleming, 1881–1955, Professor of Bacteriology, St Mary's Hospital, London, England, discovered *Penicillium notatum* in 1928.
Howard Walter Florey (Lord Florey of Adelaide), 1898–1968, Professor of Pathology, the University of Oxford, Oxford, England.
Sir Ernst Boris Chain, Professor of Biochemistry, Imperial College, London, England. Fleming, Florey and Chain shared the 1945 Nobel Prize in Physiology or Medicine for their work on penicillin.
Hans Christian Joachim Gram, 1853–1938, Professor of Pharmacology (1891–1900) and of Medicine (1900–1923), Copenhagen, Denmark, described this method of staining bacteria in 1884.

this, it is common practice in many countries to delay wound closure in patients in whom the wound is known to be contaminated or dirty. Waiting for the wound to granulate and then performing a delayed primary or secondary closure may be considered a better option in such cases (*Summary box 5.2*).

Summary box 5.2

Advances in the control of infection in surgery

- Aseptic operating theatre techniques have enhanced the use of antiseptics
- Antibiotics have reduced postoperative infection rates after elective and emergency surgery
- Delayed primary, or secondary, closure remains useful in heavily contaminated wounds

SSI in patients who have contaminated wounds, who are immunosuppressed or who are undergoing prosthetic surgery is now the exception rather than the rule since the introduction of prophylactic antibiotics. The evidence for this is of the highest level. The use of prophylactic antibiotics in clean, non-prosthetic surgery is of less value as infection rates are low and the indiscriminate use of antibiotics simply encourages the emergence of resistant strains of bacteria.

MICROBIOLOGY OF SURGICAL INFECTION

Common bacteria involved in surgical infections

Streptococci

Streptococci form chains and are Gram positive on staining (*Figure 5.2a*). The most important is the β-haemolytic *Streptococcus*, which resides in the pharynx of 5–10% of the population. In the Lancefield A–G carbohydrate antigens classification, it is the group A *Streptococcus*, also called *Streptococcus pyogenes*, that is the most pathogenic. It has the ability to spread, causing cellulitis, and to cause tissue destruction through the release of enzymes such as streptolysin, streptokinase and streptodornase.

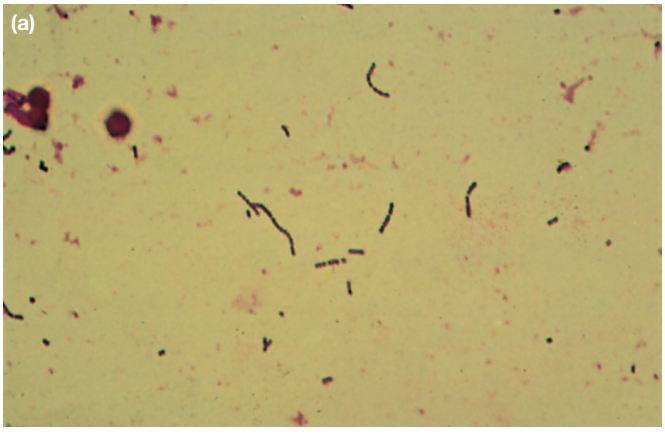
Streptococcus faecalis is an enterococcus in Lancefield group D. It is often found in synergy with other organisms, as are the γ-haemolytic *Streptococcus* and *Peptostreptococcus*, which is an anaerobe.

Both *Streptococcus pyogenes* and *Streptococcus faecalis* may be involved in wound infection after bowel surgery, but the α-haemolytic *Streptococcus viridans* is not associated with wound infections.

All the streptococci remain sensitive to penicillin and erythromycin. The cephalosporins are a suitable alternative in patients who are allergic to penicillin.

Staphylococci

Staphylococci form clumps and are Gram positive (*Figure 5.2b*). *Staphylococcus aureus* is the most important pathogen in this group and is found in the nasopharynx of up to 15% of the population. It can cause suppuration in wounds and around implanted prostheses. Some strains are resistant to many common antibiotics (especially MRSA) and so are difficult to treat. MRSA can be found in the nose of asymptomatic carriers among both patients and hospital workers, a potential source of infection after surgery. In parts of northern Europe, the prevalence of MRSA infections has been kept at very low levels using 'search and destroy' methods, which use screening techniques to look for MRSA in patients before they come in to hospital for elective surgery so that any carriers can be treated before their admission for surgery. Local policies on the management of MRSA depend on the prevalence of MRSA, the type of hospital, the clinical specialty and the availability of facilities. Widespread swabbing, ward closures, isolation of patients and disinfection of wards by deep cleaning must all be carefully considered.



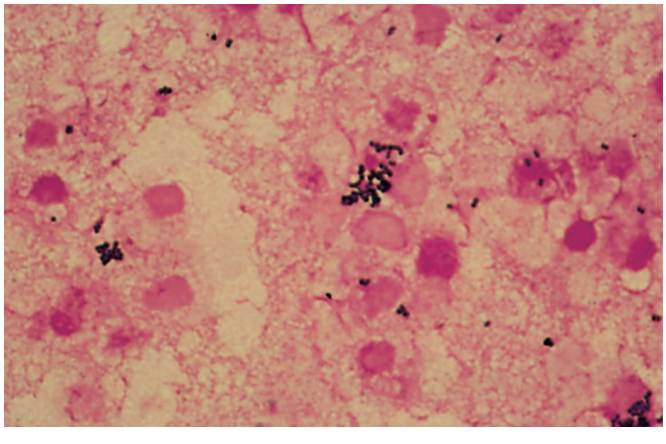


Figure 5.2 **(a)** Streptococci. **(b)** Staphylococcal pus.

Rebecca Graighill Lancefield, 1895–1981, American bacteriologist, classified streptococci in 1933.

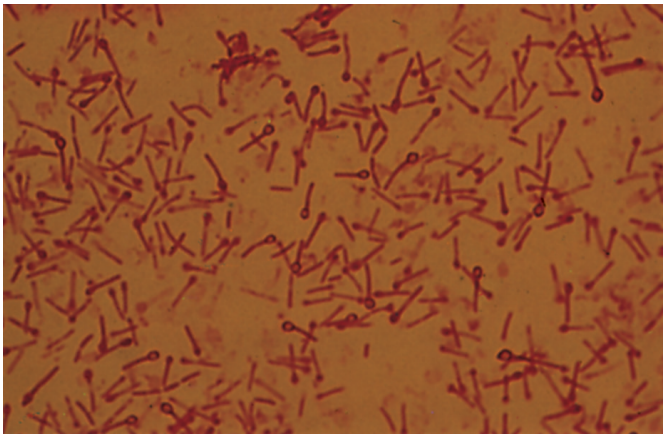


Figure 5.3 *Clostridium tetani* (drumstick spores).

Staphylococcal infections are usually suppurative and localised. Most hospital *Staphylococcus aureus* strains are now β-lactamase producers and so are resistant to penicillin, but many strains remain sensitive to flucloxacillin, vancomycin, aminoglycosides and some cephalosporins. Nowadays, several novel and innovative antibiotics have become available that have high activity against resistant strains. Some have the advantage of good oral activity (linezolid), some have a wide spectrum (teicoplanin), some have good activity in bacteraemia (daptomycin) but all are relatively expensive, and some have side effects involving marrow, hepatic and renal toxicity. Their use is justified but needs to be controlled by tight local policies and guidelines that involve clinical microbiologists.

Staphylococcus epidermidis (previously *Staphylococcus albus*), also known as coagulase-negative *Staphylococcus*, was regarded as a non-pathogenic commensal organism commonly found on the skin, but is now recognised as a major threat in vascular and orthopaedic prosthetic surgery and in indwelling vascular cannulae/catheters. The bacteria form biofilms that adhere to prosthetic surfaces and limit the effectiveness of antibiotics.

Clostridia

Clostridial organisms are Gram-positive, obligate anaerobes that produce resistant spores (***Figure 5.3***). *Clostridium perfringens* is the cause of gas gangrene, and *Clostridium tetani* causes tetanus.

Clostridium difficile (*C. diff.*) is the cause of pseudomembranous colitis, in which destruction of the normal colonic bacterial flora by antibiotic therapy allows an overgrowth of the normal gut commensal *C. diff.* to pathological levels. Any antibiotic may cause this phenomenon, although the quinolones such as ciprofloxacin seem to be the highest risk, especially in elderly or immunocompromised patients. In its most severe form, the colitis may lead to perforation and the need for emergency colectomy with an associated high mortality. Treatment involves resuscitation and antibiotic therapy. The fibrinous exudate is typical and differentiates the colitis from other inflammatory diseases.

Rapid testing for *C. diff.* is now available in the form of testing either *C. diff.* glutamate dehydrogenase (GDH) antigen or *C. diff.* toxin A/B. The *Clostridium difficile* GDH Ag Rapid test qualitatively detects for the presence of *C. diff.* GDH antigen in faeces. On the other hand, the *Clostridium difficile* Toxin A/B rapid test qualitatively detects for the presence of *C. diff.* toxins A and B in faeces. These rapid tests apply lateral flow immunochromatography and are for professional *in vitro* diagnostic use. Results are usually returned from this rapid testing in less than 30 minutes. Empirical treatment with metronidazole or vancomycin is recommended while awaiting results.

Aerobic Gram-negative bacilli

These bacilli are normal inhabitants of the large bowel. *Escherichia coli* and *Klebsiella* spp. are lactose fermenting; *Proteus* is non-lactose fermenting. Most organisms in this group act in synergy with *Bacteroides* to cause SSIs after bowel operations (in particular, appendicitis, diverticulitis and peritonitis). *Escherichia coli* is a major cause of urinary tract infection, although most aerobic Gram-negative bacilli can be involved, particularly in relation to urinary catheterisation. There is increasing concern about the development of ESBLs in many of this group of bacteria, which confer resistance to many antibiotics, particularly cephalosporins.

Pseudomonas spp. tend to colonise burns and tracheostomy wounds, as well as the urinary tract. Once *Pseudomonas* has colonised wards and intensive care units (ICUs), it may be difficult to eradicate. Surveillance of cross-infection is important in outbreaks. Hospital strains become resistant to β-lactamase as resistance can be transferred by plasmids. Wound infections need antibiotic therapy only when there is progressive or spreading infection with systemic signs. The aminoglycosides and the quinolones are effective, but some cephalosporins and penicillin may not be. Many of the carbapenems (e.g. meropenem) are useful in severe infections.

Bacteroides

Bacteroides are non-spore-bearing, strict anaerobes that colonise the large bowel, vagina and oropharynx. *Bacteroides fragilis* is the principal organism that acts in synergy with aerobic Gram-negative bacilli to cause SSIs, including intra-abdominal abscesses after colorectal or gynaecological surgery. They are sensitive to the imidazoles (e.g. metronidazole) and some cephalosporins (e.g. cefotaxime).

Sources of infection

The infection of a wound can be defined as the invasion of organisms into tissues following a breakdown of local and systemic host defences, leading to cellulitis, lymphangitis, abscess formation or bacteraemia. The infection of most surgical wounds is referred to as superficial surgical site infection (SSSI). The other categories include deep SSI (infection in the deeper musculofascial layers) and organ space infection (such

Theodor Escherich, 1857–1911, Professor of Paediatrics, Vienna, Austria, discovered the Bacterium coli commune in 1886.
Theodor Albrecht Edwin Klebs, 1834–1913, Professor of Bacteriology successively at Prague, Czechoslovakia, Zurich, Switzerland and The Rush Medical College, Chicago, IL, USA.

as an abdominal abscess after an anastomotic leak or pelvic abscess after a perforated appendicitis).

Pathogens resist host defences by releasing toxins, which favour their spread, and this is enhanced in anaerobic or frankly necrotic wound tissue. *Clostridium perfringens*, which is responsible for gas gangrene, releases proteases such as hyaluronidase, lecithinase and haemolysin, which allow it to spread through the tissues. Resistance to antibiotics can be acquired by previously sensitive bacteria by transfer through plasmids.

The human body harbours approximately 10^{14} organisms. They can be released into tissues before, during or after surgery, contamination being most severe when a hollow viscus perforates (e.g. faecal peritonitis following a diverticular perforation). Any infection that follows surgery may be termed endogenous or exogenous, depending on the source of the bacterial contamination. Endogenous organisms are present on or in the patient at the time of surgery, whereas exogenous organisms come from outside the patient. In modern hospital practice, endogenous organisms colonising the patient are by far the most common source of infection (***Summary box 5.3***).

Summary box 5.3

Classification of sources of infection

- **Endogenous**: present in or on the host, e.g. SSSI following contamination of the wound from a perforated appendix
- **Exogenous**: acquired from a source outside the body, such as the operating theatre (inadequate air filtration, poor antisepsis) or the ward (e.g. poor handwashing compliance). The cause of hospital-acquired infection (HAI)

Microorganisms are normally prevented from causing infection in tissues by intact epithelial surfaces, most notably the skin. These surfaces are broken down by trauma or surgery. In addition to these mechanical barriers, there are other protective mechanisms, which can be divided into:

- **chemical**: low gastric pH;
- **humoral**: antibodies, complement and opsonins;
- **cellular**: phagocytic cells, macrophages, polymorphonuclear cells and killer lymphocytes.

All of these natural mechanisms may be compromised by surgical intervention and treatment.

The chance of developing an SSI after surgery is also determined by the pathogenicity of the organisms present and by the size of the bacterial inoculum. The more virulent the organism or the larger the extent of bacterial contamination, the more likely is wound infection to occur. Host factors are also important, so a less virulent organism or a lower level of wound contamination may still result in a wound infection if the host response is impaired (***Summary box 5.4***). Devitalised tissue, excessive dead space or haematoma, which are all the results of poor surgical technique, increase the chances of infection. The same applies to foreign materials of any kind, including sutures and drains. These principles are important to an understanding of how best to prevent infection in surgical practice (***Summary box 5.5***).

Summary box 5.4

Factors that determine whether a wound will become infected

- Host response
- Virulence and inoculum of infective agent
- Vascularity and health of tissue being invaded (including local ischaemia as well as systemic shock)
- Presence of dead or foreign tissue
- Presence of antibiotics during the 'decisive period'

Summary box 5.5

Risk factors for increased risk of wound infection

- Malnutrition (obesity, weight loss)
- Metabolic disease (diabetes, uraemia, jaundice)
- Immunosuppression (cancer, acquired immunodeficiency syndrome [AIDS], steroids, chemotherapy and radiotherapy)
- Colonisation and translocation in the gastrointestinal tract
- Poor perfusion (systemic shock or local ischaemia)
- Foreign body material
- Poor surgical technique (devitalised tissue, dead space, haematoma)

The decisive period

There is up to a 4-hour interval before bacterial growth becomes established enough to cause an infection after a breach in the tissues, whether caused by trauma or by surgery. This interval is called the 'decisive period' and strategies aimed at preventing infection from taking a hold become ineffective after this time period. It is therefore logical that prophylactic antibiotics should be given to cover this period and that they could be decisive in preventing an infection from developing before bacterial growth takes a hold. The tissue levels of antibiotics during the period when bacterial contamination is likely to occur should be above the minimum inhibitory concentration (MIC90) for the expected pathogens.

Reduced resistance to infection

Reduced resistance to infection has several causes, particularly those that impair the inflammatory response. Host response is weakened by malnutrition associated with a low or high body mass index. Metabolic diseases such as diabetes mellitus, uraemia and jaundice, disseminated malignancy and acquired immunodeficiency syndrome (AIDS) are other contributors to infection and a poor healing response, as are iatrogenic causes including the immunosuppression caused by radiotherapy, chemotherapy and drugs such as steroids and methotrexate (***Figures 5.4 and 5.5***).

When enteral feeding is suspended during the perioperative period, and particularly with underlying disease such as cancer, immunosuppression, shock or sepsis, bacteria (particularly aerobic Gram-negative bacilli) tend to colonise the normally sterile upper gastrointestinal tract. They may then translocate

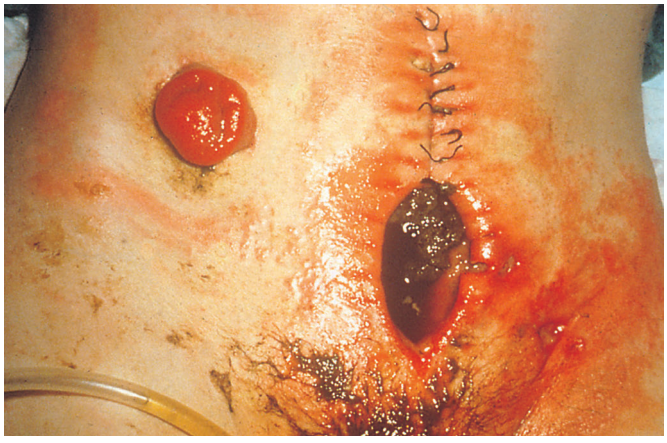


Figure 5.4 Major wound infection and delayed healing presenting as a faecal fistula in a patient with Crohn's disease on steroid treatment.

Figure 5.5 Delayed healing relating to infection in a patient on high-dose steroids.

to the mesenteric nodes and cause the release of endotoxins (lipopolysaccharide in bacterial cell walls), which can be one cause of a harmful systemic inflammatory response through the excessive release of proinflammatory cytokines and activation of macrophages (*Figure 5.6*). In the circumstances of reduced host resistance to infection, microorganisms that are not normally pathogenic may start to behave as pathogens. This is known as opportunistic infection. Opportunistic infection with fungi is an example, particularly when prolonged and changing antibiotic regimes have been used.

PRESENTATION OF SURGICAL INFECTION

Major and minor surgical site infection (SSI)

Infection acquired from the environment or the staff following surgery or admission to hospital is termed hospital-acquired infection (HAI). There are four main groups: respiratory infections (including ventilator-associated pneumonia), urinary tract infections (mostly related to urinary catheters),

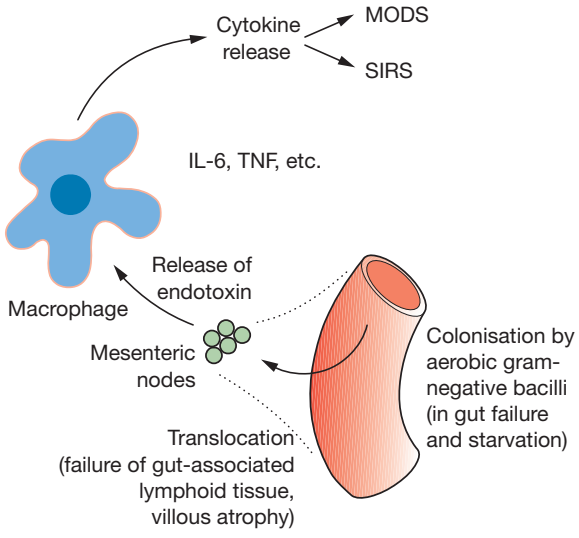


Figure 5.6 Gut failure, colonisation and translocation related to the development of systemic inflammatory response syndrome (SIRS) and multiple organ dysfunction syndrome (MODS). IL, interleukin; TNF, tumour necrosis factor.

bacteraemia (mostly related to indwelling vascular catheters) and SSIs.

A major SSI is defined as a wound that either discharges significant quantities of pus spontaneously or needs a secondary procedure to drain it (*Figure 5.4*). The patient may have systemic signs such as tachycardia, pyrexia and a raised white cell count (*Summary box 5.6*).

Summary box 5.6

Major wound infections

- Significant quantity of pus
- Delayed return home
- Patients are systemically ill

Minor wound infections may discharge pus or infected serous fluid but are not associated with excessive discomfort, systemic signs or delay in return home (*Figure 5.7*).

Figure 5.7 Minor wound infection that settled spontaneously without antibiotics.

Localised infection

Abscess

An abscess presents all the clinical features of acute inflammation originally described by Celsus: *calor* (heat), *rubor* (redness), *dolor* (pain) and *tumor* (swelling). To these can be added *functio laesa* (loss of function: if it hurts, the infected part is not used). Abscesses usually follow a puncture wound of some kind, which may have been forgotten, as well as surgery, but can be metastatic in all tissues following bacteraemia.

Pyogenic organisms, predominantly *Staphylococcus aureus,* cause tissue necrosis and suppuration. Pus is composed of dead and dying white blood cells, predominantly neutrophils, that have succumbed to bacterial toxins. An abscess is surrounded by an acute inflammatory response composed of a fibrinous exudate, oedema and the cells of acute inflammation. Granulation tissue (macrophages, fibroblasts and new blood vessel proliferation) forms later around the suppurative process and leads to collagen deposition. If it is not drained or resorbed completely, a chronic abscess may result. If it is partly sterilised with empirical antibiotics, an antibioma may form.

Abscesses contain hyperosmolar material that draws in fluid. This increases the pressure and causes pain. If they spread, they usually track along planes of least resistance and point towards the skin. Wound abscesses may discharge spontaneously by tracking to a surface but may need drainage through a surgical incision. Most abscesses relating to surgical wounds take 7–10 days to form after surgery. As many as 75% of SSIs present after the patient has left hospital and may thus be overlooked by the surgical team.

Abscess cavities need cleaning out after incision and drainage and are traditionally encouraged to heal by secondary intention. When the cavity is left open to drain freely, there is no need for antibiotic therapy as well. Antibiotics should be used if the abscess cavity is closed after drainage, but the cavity should not be closed if there is any risk of retained loculi or foreign material. Thus a perianal abscess can be incised and drained, the walls curetted and the skin closed with good results using appropriate antibiotic therapy, but a pilonidal abscess has a higher recurrence risk after such treatment because a nidus of hair may remain in the subcutaneous tissue adjacent to the abscess. Some small breast abscesses can be managed by simple needle aspiration of the pus and antibiotic therapy (***Summary box 5.7***).

Persistent chronic abscesses may lead to sinus or fistula formation. In a chronic abscess, lymphocytes and plasma cells are seen. There is tissue sequestration and later calcification may occur. Certain organisms are associated with chronicity and with sinus and fistula formation. Common ones are *Mycobacterium* and *Actinomyces.* They should not be forgotten when these complications occur and persist.

Perianastomotic contamination may be the cause of an abscess but, in the abdomen, abscesses are more usually the result of anastomotic leakage. An abscess in a deep cavity such as the pleura or peritoneum may be difficult to diagnose or locate even when there is strong clinical suspicion that it is present (***Figure 5.8***). Plain or contrast radiographs may not be helpful, but ultrasonography, computed tomography (CT), magnetic resonance imaging (MRI) and isotope-labelled white cell scans are all useful and may allow image-guided aspiration and drainage of intra-abdominal abscesses without the need for surgical intervention.

Summary box 5.7

Abscesses

- Abscesses need drainage
- Modern imaging techniques may allow guided needle aspiration, e.g. ultrasound-guided drainage of breast abscesses
- Antibiotics are indicated if the abscess cavity is not left open to drain freely
- An open abscess cavity heals by secondary intention

Cellulitis and lymphangitis

Cellulitis is a non-suppurative, invasive infection of tissues, which is usually related to the point of injury. There is poor localisation in addition to the cardinal signs of spreading inflammation. Such infections presenting in surgical practice are typically caused by organisms such as β-haemolytic streptococci (***Figure 5.9***), staphylococci (***Figure 5.10***) and *Clostridium perfringens.* Tissue destruction, gangrene and ulceration may follow, which are caused by release of proteases.

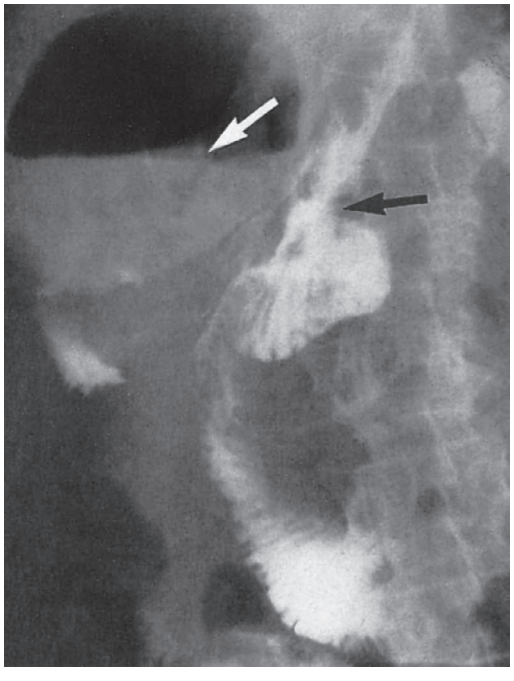


Figure 5.8 Plain radiograph showing a subphrenic abscess with a gas/fluid level (white arrow). Gastrografin is seen leaking from the oesophagojejunal anastomosis (after gastrectomy) towards the abscess (black arrow).

Aulus Aurelius Cornelius Celsus, 25 BCE–50 CE, Roman surgeon and the author of *De Re Medico Libri Octo.*

Figure 5.9 Streptococcal cellulitis of the leg following a minor puncture wound.

Figure 5.10 Staphylococcal cellulitis of the face and orbit following severe infection of an epidermoid cyst of the scalp.

Systemic signs (the old-fashioned term is toxaemia) are common, with chills, fever and rigors. These events follow the release of toxins into the circulation, which stimulate a cytokine-mediated systemic inflammatory response even though blood cultures may be negative.

Lymphangitis is part of a similar process and presents as painful red streaks in affected lymphatics draining the source of infection. Lymphangitis is often accompanied by painful lymph node groups in the related drainage area, e.g. cervical, axillary or inguinal (*Summary box 5.8*).

Summary box 5.8

Cellulitis and lymphangitis

- Non-suppurative, poorly localised
- Commonly caused by streptococci, staphylococci or clostridia
- Blood cultures are often negative

Specific local wound infections

Gas gangrene

Gas gangrene is caused by *Clostridium perfringens*. These Gram-positive, anaerobic, spore-bearing bacilli are widely found in nature, particularly in soil and faeces. Patients who are immunocompromised, diabetic or have malignant disease are at greater risk, particularly if they have wounds containing necrotic or foreign material, resulting in anaerobic conditions. Military wounds provide an ideal environment as the kinetic energy of high-velocity missiles or shrapnel causes extensive tissue damage. The cavitation which follows passage of a missile through the tissues causes a 'sucking' entry wound, leaving clothing and environmental soiling in the wound in addition to devascularised tissue. Gas gangrene wound infections are associated with severe local wound pain and crepitus (gas in the tissues, which may also be visible on plain radiographs). The wound produces a thin, brown, sweet-smelling exudate, in which Gram staining will reveal bacteria. Oedema and spreading gangrene follow the release of collagenase, hyaluronidase, other proteases and alpha toxin. Early systemic complications with circulatory collapse and organ failure follow if prompt action is not taken (*Summary box 5.9*).

Antibiotic prophylaxis should always be considered in patients at risk, especially when amputations are performed for peripheral vascular disease with open necrotic ulceration. Once gas gangrene infection is established, large doses of intravenous penicillin and aggressive debridement of affected tissues are required.

Summary box 5.9

Gas gangrene

- Caused by *Clostridium perfringens*
- Gas and smell are characteristic
- Immunocompromised patients are most at risk
- Antibiotic prophylaxis is essential when performing amputations to remove dead tissue

Clostridium tetani

This is another anaerobic, terminal spore-bearing, Gram-positive bacterium, which can cause tetanus following implantation into tissues or a wound. The spores are widespread in soil and manure. The signs and symptoms of tetanus are mediated by the release of the exotoxin tetanospasmin; these include spasms in the distribution of the short motor nerves of the face followed by the development of severe generalised motor spasms including opisthotonus, respiratory arrest and death. Prophylaxis with tetanus toxoid is the best preventative treatment but, in an established infection, minor debridement of the wound may need to be performed and antibiotic treatment with benzylpenicillin provided in addition. Relaxants may also be required and the patient will require ventilation in severe forms, which are associated with a high mortality.

Synergistic spreading gangrene (synonym: subdermal gangrene, necrotising fasciitis)

This is a rare but serious bacterial infection that affects and spreads via the deep fascia; hence, it is termed fasciitis. A mixed pattern of organisms is responsible for this serious condition: coliforms, staphylococci, *Bacteroides* spp., anaerobic streptococci and peptostreptococci have all been implicated, acting in synergy. Often, aerobic bacteria destroy the living tissue, allowing anaerobic bacteria to thrive. Severe wound pain, signs of spreading inflammation with crepitus and odour are all signs of the infection spreading. Untreated, it will lead to widespread local gangrene and systemic multisystem organ failure. Abdominal wall infections are known as Meleney's synergistic gangrene and scrotal infections as Fournier's gangrene (***Figure 5.11***). Patients are almost always immunocompromised, with conditions such as diabetes mellitus. The wound initiating the infection may have been minor, but severely contaminated wounds are more likely to be the cause.

The subdermal spread of gangrene is always much more extensive than appears from initial examination. The finger test can be used in the diagnosis of patients who present with suspected necrotising fasciitis. The area of suspected involvement is first infiltrated with local anaesthesia. A 2-cm incision is made in the skin down to the deep fascia. Lack of bleeding is a sign of necrotising fasciitis. On some occasions, a dishwater-coloured fluid is noticed seeping from the wound.

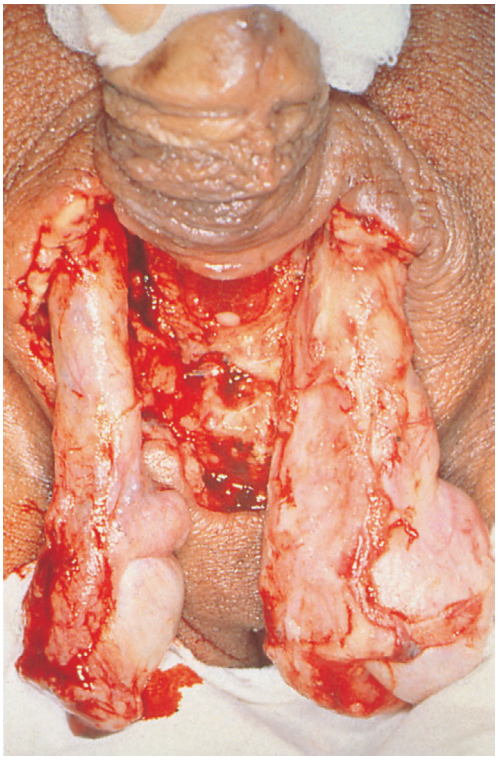


Figure 5.11 A classic presentation of Fournier's gangrene of the scrotum with 'shameful exposure of the testes' following excision of the gangrenous skin.

A gentle, probing manoeuvre with the index finger covered by a sterile glove is then performed at the level of the deep fascia. If the tissues dissect with minimal resistance, the finger test is positive. Tissue biopsies are then sent for frozen section analysis. The characteristic histological findings are obliterative vasculitis of the subcutaneous vessels, acute inflammation and subcutaneous tissue necrosis. If either the finger test or rapid frozen section analysis is positive, or if the patient has progressive clinical findings consistent with necrotising fasciitis, immediate operative treatment must be initiated.

Broad-spectrum antibiotic therapy must be combined with aggressive circulatory support. Locally, there should be wide excision of necrotic tissue and laying open of affected areas. The debridement may need to be extensive, and patients who survive may need large areas of skin grafting later.

Systemic infection

Bacteraemia

Bacteraemia is unusual following superficial SSIs, which tend to drain through the wound, but common after deep space SSIs, such as following an intestinal anastomotic breakdown. It is usually transient and can follow procedures undertaken through infected tissues (particularly instrumentation in infected bile or urine). It may also occur through bacterial infection of indwelling intravenous cannulae, which should be replaced regularly to avoid colonisation. Bacteraemia is important when a prosthesis has been implanted, as infection of the prosthesis can occur through haematogenous spread. Aerobic Gram-negative bacilli are often responsible, but *Staphylococcus aureus* and fungi may be involved (***Summary box 5.10***).

Summary box 5.10

Bacteraemia

- Common after anastomotic breakdown
- Dangerous if the patient has a prosthesis, which can become infected
- May be associated with systemic organ failure

Systemic inflammatory response syndrome

Systemic inflammatory response syndrome (SIRS) is a systemic manifestation of sepsis (***Table 5.1***), although the syndrome may also be caused by multiple trauma, burns or pancreatitis without infection. Serious infection, such as secondary peritonitis, may lead to SIRS through the release of lipopolysaccharide endotoxin from the walls of dying Gram-negative bacilli (mainly *Escherichia coli*) or other bacteria or fungi. This and other toxins stimulate the release of cytokines from macrophages (***Figure 5.6***). SIRS should not be confused with bacteraemia, although the two may coexist.

Frank Lamont Meleney, 1889–1963, Professor of Clinical Surgery, Columbia University, New York, NY, USA.
Jean Alfred Fournier, 1832–1915, syphilologist, the founder of the Venereal and Dermatological Clinic, Hôpital Saint-Louis, Paris, France.

TABLE 5.1 Definitions of systemic inflammatory response syndrome (SIRS) and sepsis

SIRS is *Presence of two out of three of the following:* • Hyperthermia (>38°C) or hypothermia (<36°C) • Tachycardia (>90/min, no β-blockers) or tachypnoea (>20/min) • White cell count >12 × 10^9/litre or <4 × 10^9/litre
Sepsis is SIRS with a documented source of infection
Severe sepsis or sepsis syndrome is sepsis with evidence of failure of one or more organs: respiratory (acute respiratory distress syndrome), cardiovascular (septic shock follows compromise of cardiac function and fall in peripheral vascular resistance), renal (usually acute tubular necrosis), hepatic, blood coagulation systems or central nervous system

Septic manifestations and multiple organ dysfunction syndrome (MODS) in SIRS are mediated by the release of pro-inflammatory cytokines such as interleukin-1 (IL-1) and tumour necrosis factor alpha (TNFα). These cytokines normally stimulate neutrophil adhesion to endothelial surfaces adjacent to the source of infection and cause them to migrate through the blood vessel wall by chemotaxis, where they can attack the bacterial invasion. A respiratory burst occurs within such activated neutrophils, releasing lysosomal enzymes, oxidants and free radicals, which are involved in killing the invading bacteria but which may also damage adjacent cells. Coagulation, complement and fibrinolytic pathways are also stimulated as part of the normal inflammatory response. This response is usually beneficial to the host and is an important aspect of normal tissue repair and wound healing. On occasions, this response may become harmful to the host if it occurs in excess, when it is known as the systemic inflammatory response syndrome or SIRS. There are high circulating levels of cytokines and activated neutrophils that stimulate fever, tachycardia and tachypnoea. The activated neutrophils adhere to vascular endothelium in key organs remote from the source of infection and damage it, leading to increased vascular permeability, which in turn leads to cellular damage within the organs, which become dysfunctional and give rise to the clinical picture of multiple organ dysfunction syndrome or MODS. In its most severe form, MODS may progress into multiple system organ failure (MSOF). Respiratory, cardiac, intestinal, renal and liver failure ensue in combination with circulatory failure and shock. In this state, the body's resistance to infection is reduced and a vicious cycle develops where the more organs that fail, the more likely it becomes that death will follow despite all that a modern ICU can do for organ support (*Summary box 5.11*).

Summary box 5.11

Definitions of infected states

- SSI is an infected wound or deep organ space
- SIRS is the body's systemic response to severe infection
- MODS is the effect that SIRS produces systemically
- MSOF is the end stage of uncontrolled MODS

Surviving Sepsis Campaign/sepsis bundle/ Sepsis Six

The European Society of Intensive Care Medicine (ESICM) alongside the Society of Critical Care Medicine (SCCM) spearheaded the Surviving Sepsis Campaign (SSC) in 2002 with several aims, including the development of guidelines for the diagnosis, treatment and post-ICU care of sepsis and a reduction in mortality from sepsis. The Surviving Sepsis Campaign continually develops and updates resources and implementation tools to further its mission of reducing sepsis and septic shock.

The **sepsis bundle**, also known as the resuscitation bundle, is a combination of evidence-based objectives that must be completed within 6 hours for patients presenting with severe sepsis, septic shock and/or lactate >4 mmol/L.

The **Sepsis Six** is the name given to a bundle of medical therapies designed to reduce mortality in patients with sepsis. Drawn from international guidelines that emerged from the Surviving Sepsis Campaign, the Sepsis Six was developed by the UK's Sepsis Trust. The components of the Sepsis Six are:

- **give three to patients:** (1) intravenous fluid challenge, (2) intravenous antibiotics, (3) oxygen and monitor urine output;
- **take three from patients:** (4) blood cultures, (5) full blood count, (6) lactate.

Viral infections relevant to surgery

Hepatitis

Both hepatitis B and hepatitis C carry risks in surgery as they are blood-borne pathogens that can be transmitted both from the patient to the surgeon and vice versa. The usual mode of transmission is blood-to-blood contact through a needle-stick injury or a cut. Many cases of hepatitis B are asymptomatic and a surgeon may carry the virus without being aware of it. As there is an effective vaccine against hepatitis B, surgeons should know their immune status to hepatitis B and be vaccinated against it. Hepatitis C infection often becomes chronic with the risk of significant liver damage but is potentially curable with interferon-alpha and ribavirin treatment, so surgeons who are exposed to an infection risk should seek medical advice and antibody measurement.

Human immunodeficiency virus

The type I human immunodeficiency virus (HIV) is one of the viruses of surgical importance because it can be transmitted by body fluids, particularly blood. It is a retrovirus that has become increasingly prevalent through sexual transmission (both homo- and heterosexual), intravenous drug addiction and in infected blood products used to treat patients with haemophilia in particular. The risk in surgery is mostly through needle-stick injury during operations.

The risk of opportunistic infections (such as *Pneumocystis carinii* pneumonia, tuberculosis and cytomegalovirus) and neoplasms (such as Kaposi's sarcoma and lymphoma) is thereby increased.

Moritz Kaposi, 1837–1902, Professor of Dermatology, Vienna, Austria, described pigmented sarcoma of the skin in 1872.

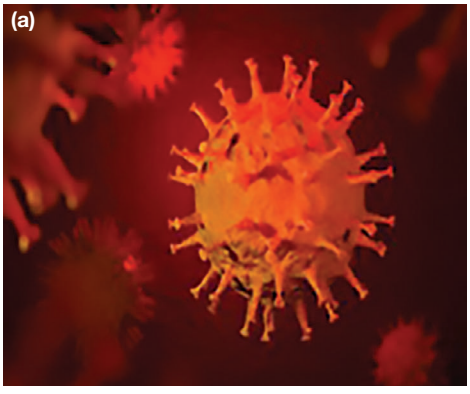


Figure 5.12 **(a)** Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). **(b)** Chest radiograph showing coronavirus disease 2019 (COVID-19) pneumonia changes.

In the early weeks after HIV infection, there may be a flu-like illness and, during the phase of seroconversion, patients present the greatest risk of HIV transmission. It is during these early phases that drug treatment, highly active antiretroviral therapy (HAART), is most effective through the ability of these drugs to inhibit reverse transcriptase and protease synthesis, which are the principal mechanisms through which HIV can progress. These drugs suppress the virus but do not clear it completely from the body and treated patients can still transmit the virus to others. Within 2 years, untreated HIV can progress to acquired immunodeficiency syndrome (AIDS) in 25–35% of patients.

Universal precautions

Patients may present to surgeons for operative treatment if they have a surgical disease and they are known to be infected or 'at risk', or because they need surgical intervention related to their illness for vascular access or a biopsy when they are known to have hepatitis, HIV infection or AIDS. Particular care should be taken when there is a risk of splashing/aerosol formation, particularly with power tools. Universal precautions have been drawn up by CDC in the USA and largely adopted by the NHS in the UK. In summary, these are:

- use of a full face mask ideally, or protective spectacles;
- use of fully waterproof, disposable gowns and drapes, particularly during seroconversion;
- boots to be worn, not clogs, to avoid injury from dropped sharps;
- double gloving needed (a larger size on the inside is more comfortable);
- allow only essential personnel in theatre;
- avoid unnecessary movement in theatre;
- respect is required for sharps, with passage in a kidney dish;
- slow meticulous operative technique is needed with minimised bleeding.

After contamination

Needle-stick injuries are commonest on the non-dominant index finger during operative surgery. Hollow needle injury carries the greatest risk of viral transmission. The injured part should be washed under running water and the incident reported. Local policies dictate whether postexposure antiretroviral treatment should be given. Occupational health advice is required after high-risk exposure, together with the need for hepatitis/HIV testing and the option for continuation in a non-operative specialty.

COVID-19 pandemic

The global pandemic of coronavirus disease 2019 (COVID-19) was announced by the World Health Organization on 11 March 2020. As of 16 August 2021, more than 207 million cases and more than 4.36 million deaths had been reported in 210 countries.* The rapid spread of the outbreak has had short-term implications for global healthcare systems, including the field of surgery. Many hospitals were forced to stop or postpone elective surgical interventions during the first wave in early to mid-2020. However, emergency surgery and time-sensitive surgery, i.e. cancer surgery, continued with relevant precautions, including the use of personal protective equipment.

COVID-19 is a contagious respiratory and vascular disease. The aetiology is severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) (***Figure 5.12a***), which is a specific type of coronavirus. Common symptoms include fever, cough, fatigue, shortness of breath or breathing difficulties as well as loss of smell and taste. The incubation period may range from 1 to 14 days. While most people have mild symptoms, some people develop acute respiratory distress syndrome (ARDS), possibly precipitated by a cytokine storm; multiorgan failure; septic shock; and hypercoagulable states. Longer term damage to organs (in particular, the lungs and heart) has been observed.

Complications in postoperative surgical patients infected with COVID-19 in either an elective or emergency setting may include pneumonia (***Figure 5.12b***), ARDS, multiorgan failure, septic shock and death. Measures to control COVID-19-related morbidity and mortality have thus been implemented by many countries, including a mandatory preoperative cocooning of elective surgical patients 10–14 days prior to surgery and preoperative COVID-19 swab testing 2–3 days prior to elective surgery. Like many other viral infections, there is no definite pharmacological cure for this infection, although there has been some evidence for supportive care, e.g. with hydroxychloroquine and dexamethasone. Multiple vaccines were made available at the end of 2020 and, currently, the majority of countries have most of their population vaccinated.

*Data from: COVID-19 dashboard by the Center for Systems Science and Engineering (CSSE) at Johns Hopkins University (accessed 16 August 2021).

PREVENTION OF SURGICAL INFECTION

Preoperative preparation

A short preoperative hospital stay lowers the risk of acquiring MRSA, multidrug-resistant coagulase-negative staphylococci and other antibiotic-resistant organisms from the hospital environment. Medical and nursing staff should always wash their hands after any patient contact. Hand gels containing at least 70% alcohol can act as a substitute for handwashing, but do not destroy the spores of *C. diff.*, which may cause pseudomembranous colitis, especially in immunocompromised patients or those whose gut flora is suppressed by antibiotic therapy. Although the need for clean hospitals, emphasised by the media, is logical, the 'clean your hands campaign', particularly in the COVID-19 era, is beginning to result in falls in the incidence of HAIs. Staff with open, infected skin lesions should not enter the operating theatres. Ideally, neither should affected patients, especially if they are having a prosthesis implanted. Antiseptic baths (usually chlorhexidine) are popular in Europe, but there is no hard evidence for their value in reducing wound infections. Preoperative skin shaving should be undertaken in the operating theatre immediately before surgery as the SSI rate after clean wound surgery may be doubled if shaving is performed the night before because minor skin injury enhances superficial bacterial colonisation.

Scrubbing and skin preparation

When washing the hands prior to surgery, dilute alcohol-based antiseptic hand soaps such as chlorhexidine or povidone–iodine should be used, and the scrub should include the nails (***Figure 5.1***).

One application of a more concentrated alcohol-based antiseptic is adequate for skin preparation of the operative site. This leads to a >95% reduction in bacterial count but caution should be taken not to leave a pool of alcohol-based fluid on the skin as it could ignite with diathermy and burn the patient (***Figure 5.1***).

Theatre technique and discipline also contribute to low infection rates. Numbers of staff in the theatre and movement in and out of theatre should be kept to a minimum. Careful and regular surveillance is needed to ensure the quality of instrument sterilisation, aseptic technique and theatre ventilation. Laminar flow systems direct clean, filtered air over the operating field, with any air potentially contaminated as it passes over the incision then directed away from the patient. Operator skill in gentle manipulation and dissection of tissues is much more difficult to audit, but dead spaces and haematomas should be avoided. There is no evidence that drains, incision drapes or wound guards help to reduce wound infection. There is a high level of evidence that both the perioperative avoidance of hypothermia and the use of supplemental oxygen during recovery significantly reduce the rate of SSIs.

Prophylactic antibiotics

Prophylactic antibiotics are used when there is a risk of wound contamination with bacteria during surgery. The theoretical degree of contamination, proposed by the National Research Council (USA) over 40 years ago, relates well to infection rates (***Table 5.2***). The value of antibiotic prophylaxis is low in non-prosthetic clean surgery, with most trials showing no clear benefit because infection rates without antibiotics are so low. The exception to this is where a prosthetic implant is used, as the results of infection are so catastrophic that even a small risk of infection is unacceptable. There is undisputed evidence that prophylactic antibiotics are effective in reducing the risk of infection in clean-contaminated and contaminated operations. When wounds are heavily contaminated or when an incision is made into an abscess, a 5-day course of therapeutic antibiotics may be justified on the assumption that the wound is inevitably infected and so treatment is needed rather than prophylaxis.

TABLE 5.2 Surgical site infection rates relating to wound contamination with and without using antibiotic prophylaxis

Type of surgery	Infection rate with prophylaxis (%)	Infection rate without prophylaxis (%)
Clean (no viscus opened)	1–2	1–2
Clean-contaminated (viscus opened, minimal spillage)	3	6–9
Contaminated (open viscus with spillage or inflammatory disease)	6	13–20
Dirty (pus or perforation, or incision through an abscess)	7	40

If antibiotics are given to prevent infection after surgery or instrumentation, they should be used before bacterial growth becomes established (i.e. within the decisive period). Ideally, maximal blood and tissue levels should be present at the time at which the first incision is made and before contamination occurs. Tissue levels of the antibiotic should remain high throughout the operation and antibiotics with a short tissue half-life should be avoided. Intravenous administration at induction of anaesthesia is therefore optimal, as unexpected delays in the timing of surgery may occur before then and antibiotic tissue levels may fall off before the surgery starts. In long operations or when there is excessive blood loss, or when unexpected contamination occurs, antibiotics may be repeated at 4-hourly intervals during the surgery because tissue antibiotic levels often fall faster than serum levels. There is no evidence that further doses of antibiotics after surgery are of any value in prophylaxis against infection and the practice can only encourage the development of antibiotic resistance. The choice of an antibiotic depends on the expected spectrum of organisms likely to be encountered, which will depend on the site and type of surgery and whether the patient has any antibiotic allergies. Hospitals in the UK and across Europe now have standardised antibiotic prophylaxis policies that take account of the above factors and are only deviated from with microbiological advice.

Patients with known valvular disease of the heart (or with any implanted vascular or orthopaedic prosthesis) should have prophylactic antibiotics during dental, urological or open

viscus surgery to prevent bacterial colonisation of the valve or prosthesis during the transient bacteraemia which can occur during such surgery (*Summary box 5.12*).

Summary box 5.12

Antibiotic prophylaxis

- Not required in clean surgery unless a prosthesis is implanted
- Use antibiotics that are effective against expected pathogens within local hospital guidelines
- Plan for single-shot intravenous administration at induction of anaesthesia
- Repeat only during long operations or if there is excessive blood loss
- Patients with heart valve disease or a prosthesis should be protected from bacteraemia caused by dental work, urethral instrumentation or visceral surgery

Postoperative wound infections

The majority of wound infections arise from endogenous sources within the patient, but exogenous SSIs may also occur from bacteria present in the ward or on staff and so can be related to poor hospital standards. Strict attention to ward cleanliness, gloving before touching patient wounds and hand-washing between all patient contacts are important preventive measures. An outbreak of wound infections on the ward with bacteria having the same antibiotic sensitivity profile implies an exogenous source of infection, which needs to be investigated by swabbing all staff and work surfaces. It may need temporary ward closure and a deep clean to eradicate the infection source.

Now that patients are discharged more quickly after surgery and many procedures are performed as day cases, many SSIs are missed by the surgical team unless they undertake a prolonged and carefully audited follow-up with primary care doctors. Suppurative wound infections take 7–10 days to develop, and even cellulitis around wounds caused by invasive organisms (such as β-haemolytic *Streptococcus*) takes 3–4 days to develop. Major surgical infections with systemic signs (*Figure 5.13*), evidence of spreading infection, cellulitis or bacteraemia need treatment with appropriate antibiotics. The choice may need to be empirical initially but is best based on culture and sensitivities of isolates harvested at surgery or from culture of wound fluids or wound swabs. Although the identification of organisms in surgical infections is necessary for audit and wound surveillance purposes, it is usually 2–3 days before sensitivities are known (*Figures 5.14 and 5.15*). It is illogical to withhold antibiotics until results are available but, if clinical response is poor by the time sensitivities are known, then antibiotics can be changed. Such changes are unusual if the empirical choice of antibiotics is sensible; change of antibiotics promotes resistance and risks complications, such as *C. diff.* enteritis.

If an infected wound is under tension, or there is clear evidence of suppuration, sutures or clips need to be removed, with curettage if necessary, to allow pus to drain adequately. In severely contaminated wounds, such as an incision made for drainage of an abscess, it is logical to leave the skin open. Delayed primary or secondary closure can be undertaken when the wound is clean and granulating (*Figures 5.16 and 5.17*). Some heavily infected wounds may be left to heal by secondary intention, with no attempt at closure, particularly where there is a loss of skin cover and healthy granulation tissue develops

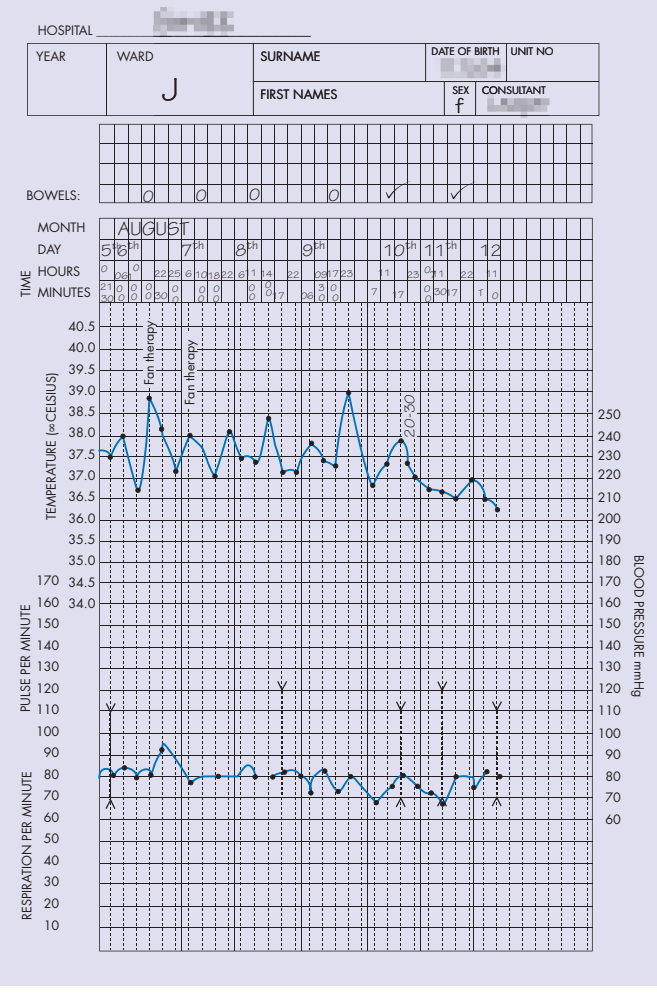


Figure 5.13 Classic swinging pyrexia related to a perianastomotic wound abscess that settled spontaneously on antibiotic therapy.

Figure 5.14 Mixed streptococcal infection of a skin graft with very poor 'take'.

Figure 5.15 After 5–6 days of antibiotics, the infection shown in *Figure 5.14* is under control, and the skin grafts are clearly viable.

Figure 5.16 (a, b) Delayed primary closure of a fasciotomy wound after 3–5 days.

(*Figure 5.18*). While the end result may be excessive scarring, that can always be revised with plastic surgery under clean surgical conditions at a later stage (*Summary box 5.13*).

Summary box 5.13

Surgical incisions through infected or contaminated tissues

- When possible, tissue or pus for culture should be taken before antibiotic cover is started
- The choice of antibiotics is empirical until sensitivities are available
- Heavily contaminated wounds are best managed by delayed primary or secondary closure

When taking pus from infected wounds, specimens should be sent fresh for microbiological culture. Swabs should be placed in transport medium, but the larger the volume of pus sent, the more likely is the accurate identification of the organism involved. Providing the microbiologist with as much information as possible and discussing the results with them gives the best chance of the most appropriate antibiotic treatment. If bacteraemia is suspected, but results are negative, then repeat specimens for blood culture and an immediate Gram stain.

Topical antiseptics should only be used on heavily contaminated wounds for a short period to clear infection as they inhibit epithelial ingrowth and so impair wound healing.

ANTIMICROBIAL TREATMENT OF SURGICAL INFECTION

Principles

Antimicrobials may be used to prevent or treat established surgical infection.

The use of antibiotics for the treatment of established surgical infection ideally requires recognition and determination of the sensitivities of the causative organisms. Antibiotic therapy

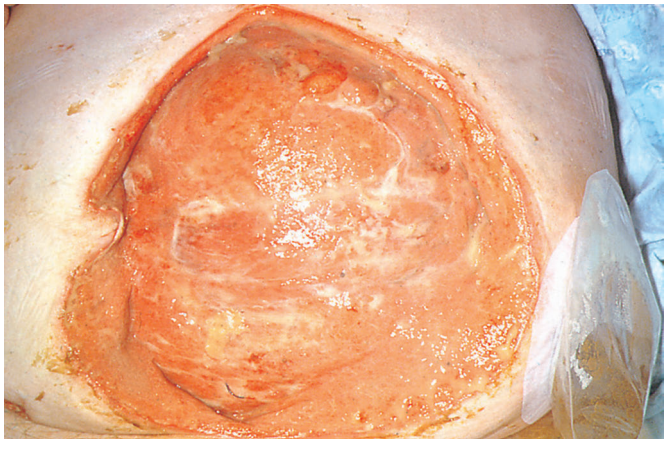


Figure 5.17 Skin layers left open to granulate after laparotomy for faecal peritonitis, ready for skin grafting.

Figure 5.18 Infected animal bite/wound of the upper thigh, treated by open therapy following virulent staphylococcal infection.

should not be held back if it is indicated, the choice being empirical and later modified depending on microbiological findings on culture and sensitivity. Once antibiotics have been administered, it may not be possible to grow bacteria from the wound and so the opportunity to ascertain the most appropriate antibiotic sensitivities is lost if a patient's condition does not improve on empirical antibiotic therapy. Antibiotics alone are rarely sufficient to treat SSIs, which may also need open drainage and debridement (*Summary box 5.14*).

Summary box 5.14

Principles for the use of antibiotic therapy

- Antibiotics do not replace surgical drainage of infection
- Only spreading infections or signs of systemic infection justify the use of antibiotics
- Whenever possible, the organism and sensitivity should be determined

There are two approaches to antibiotic treatment:

A **narrow-spectrum antibiotic** may be used to treat a known sensitive infection; for example, MRSA (which may be isolated from pus) is usually sensitive to vancomycin or teicoplanin, but not flucloxacillin.

Combinations of broad-spectrum antibiotics can be used when the organism is not known or when it is suspected that several bacteria, acting in synergy, may be responsible for the infection. For example, during and following emergency surgery requiring the opening of perforated or ischaemic bowel, any of the gut organisms may be responsible for subsequent peritoneal or bacteraemic infection. In this case, a broad-spectrum antibiotic such as teicoplanin or meropenem, which are effective against a wide range of aerobic bacteria, is combined with metronidazole, which is effective against anaerobic bacteria. Alternatively, triple therapy is used with amoxicillin, gentamicin and metronidazole. The use of such broad-spectrum antibiotic strategies should be guided by specialist microbiological advice. If clinical response is poor after 3–4 days, there should be a re-evaluation with a review of available culture and sensitivity results and further investigations requested to exclude the development or persistence of infection such as a collection of pus.

In surgical units in which resistant *Pseudomonas* or other Gram-negative species (such as *Klebsiella*) have become 'resident opportunists', it may be necessary to rotate anti-pseudomonal and anti-Gram-negative antibiotic therapy (*Summary box 5.15*).

Summary box 5.15

Treatment of commensals that have become opportunist pathogens

- They are likely to have multiple antibiotic resistance
- It may be necessary to rotate antibiotics

Antibiotics used in treatment and prophylaxis of surgical infection

Antimicrobials may be produced by living organisms (antibiotics) or by synthetic methods. Some are bactericidal, e.g. penicillins and aminoglycosides, and others are bacteriostatic, e.g. tetracycline and erythromycin. In general, penicillins act upon the bacterial cell wall and are most effective against bacteria that are multiplying and synthesising new cell wall materials. The aminoglycosides act at the ribosomal level, preventing or distorting the production of proteins required to maintain the integrity of the enzymes in the bacterial cell. Hospital and formulary guidelines should be consulted for doses and monitoring of antibiotic therapy.

Penicillin

Benzylpenicillin has proved most effective against Gram-positive pathogens, including most streptococci, the clostridia and some of the staphylococci that do not produce β-lactamase. It is still effective against *Actinomyces,* which is a rare cause of chronic wound infection. It may be used specifically to treat spreading streptococcal infections. Penicillin is valuable even if other antibiotics are required as part of multiple therapy for a mixed infection. Some serious infections, e.g. gas gangrene, require high-dose intravenous benzylpenicillin.

Flucloxacillin

Flucloxacillin is resistant to β-lactamases, and is therefore of use in treating infections with penicillinase-producing staphylococci that are resistant to benzylpenicillin, but it has poor activity against other pathogens. It has good tissue penetration and therefore is useful in treating soft-tissue infections and osteomyelitis.

Ampicillin, amoxicillin and co-amoxiclav

Ampicillin and amoxicillin are β-lactam penicillins and can be taken orally or may be given parenterally. Both are effective against Enterobacteriaceae, *Enterococcus faecalis* and the majority of group D streptococci, but not species of *Klebsiella* or *Pseudomonas.* Clavulanic acid has no antibacterial activity itself, but it does inactivate β-lactamases, so can be used in conjunction with amoxicillin. The combination is known as co-amoxiclav and is useful against β-lactamase-producing bacteria that are resistant to amoxicillin on its own. These include resistant strains of *Staphylococcus aureus, Escherichia coli, Haemophilus influenzae, Bacteroides* and *Klebsiella.*

Piperacillin and ticarcillin

These are ureidopenicillins with a broad spectrum of activity against a broad range of Gram-positive, Gram-negative and anaerobic bacteria. Both are used in combination with β-lactamase inhibitors (tazobactam with piperacillin and clavulanic acid with ticarcillin). They are not active against MRSA but are used in the treatment of septicaemia, hospital-acquired pneumonia and complex urinary tract infections, where they are active against *Pseudomonas* and *Proteus* spp. and have a synergistic effect when used with aminoglycosides such as gentamicin.

Cephalosporins

There are several β-lactamase-susceptible cephalosporins that are of value in surgical practice: cefuroxime, cefotaxime and ceftazidime are widely used. The first two are most effective in intra-abdominal skin and soft-tissue infections, being active against *Staphylococcus aureus* and most Enterobacteriaceae. As a group, the enterococci (*Streptococcus faecalis*) are not sensitive to the cephalosporins. Ceftazidime, although active against the Gram-negative organisms and *Staphylococcus aureus*, is also effective against *Pseudomonas aeruginosa*. These cephalosporins may be combined with an aminoglycoside, such as gentamicin, if Gram-negative cover is needed, and an imidazole, such as metronidazole, if anaerobic cover is needed.

Aminoglycosides

Gentamicin and tobramycin have similar activity and are effective against Gram-negative Enterobacteriaceae. Gentamicin is effective against many strains of *Pseudomonas*, although resistance has been recognised. All aminoglycosides are inactive against anaerobes and streptococci. Serum levels immediately before and 1 hour after intramuscular injection must be taken and repeated at 48 hours after the start of therapy, and dosage should be modified to satisfy peak and trough levels. Ototoxicity and nephrotoxicity may follow sustained high toxic levels and therefore single, large doses may be safer.

Vancomycin and teicoplanin

These glycopeptide antibiotics are most active against Gram-positive aerobic and anaerobic bacteria and have proved to be effective against MRSA, so are often used as prophylactic antibiotics when there is a high risk of MRSA. They are ototoxic and nephrotoxic, so serum levels should be monitored. They are effective against *C. diff.* in cases of pseudomembranous colitis.

Carbapenems

Meropenem, ertapenem and imipenem are members of the carbapenems. They are stable to β-lactamase, have useful broad-spectrum anaerobic as well as Gram-positive activity and are effective for the treatment of resistant organisms, such as ESBL-resistant urinary tract infections or serious mixed-spectrum abdominal infections (peritonitis).

Metronidazole

Metronidazole is the most widely used member of the imidazole group and is active against all anaerobic bacteria. It is particularly safe and may be administered orally, rectally or intravenously. Infections caused by anaerobic cocci and strains of *Bacteroides* and *Clostridia* can be treated, or prevented, by its use. Metronidazole is useful for the prophylaxis and treatment of anaerobic infections after abdominal, colorectal and pelvic surgery and in the treatment of *C. diff.* pseudomembranous colitis.

Ciprofloxacin

Quinolones, such as ciprofloxacin, have a broad spectrum of activity against both Gram-positive and Gram-negative bacteria but are particularly useful against *Pseudomonas* infections. Many UK and European hospitals have restricted their use as a preventive measure against the development of *C. diff.* enterocolitis.

FURTHER READING

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